

Building a Circuit in LogiSim

لاجی سیم میں $Y = A \cdot B$ بنانے کے لیے AND گیٹ رکھیں، بائیں جانب دو ان پٹ پن (چوکور) اور دائیں جانب ایک آؤٹ پٹ پن (...)

CS XI

Chapter 1: Computer Systems

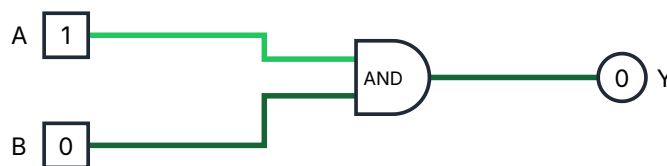
Lecture 14 of 20

Building a Circuit in LogiSim

WORKED EXAMPLE

Step by step: building $Y = A \cdot B$

1. Select the **AND gate** from the Toolbar (or from the Gates library) and place it on the canvas.
2. Select the **Input Pin** (square icon with a 0) and place two pins to the left of the gate. Label them A and B.
3. Select the **Output Pin** (circle icon with a 0) and place it to the right. Label it Y.
4. Use the **Wiring Tool** to click and drag wires from the input pins to the AND gate's input ports. Wires must be horizontal or vertical.
5. Wire the gate's output to the output pin Y.
6. Select the **Poke Tool** (pointing finger) and click inputs A and B to toggle them between 0 and 1. Test the circuit!



Bright green = 1, dark green = 0

Testing with the Poke Tool: A = 1 (bright green wire), B = 0 (dark green), so $Y = 1 \cdot 0 = 0$.

WORKED EXAMPLE

Testing the circuit = making its truth table. Poke the inputs through all four combinations and write what Y shows:

Poke A	Poke B	Y shows	Output wire colour
0	0	0	Dark green
0	1	0	Dark green
1	0	0	Dark green
1	1	1	Bright green

This matches the AND truth table, so the circuit is correct. Try the same with $Y = A \cdot B + C$ from 1.1.11 (add a second gate and a third input pin).

REMEMBER IT

Input pin = **square** (you push it). Output pin = **circle** (it shows a light). Poke = finger — you "poke" inputs to change them.

COMMON MISTAKE

Students often try to poke inputs while the Wiring or Select tool is active — then clicking moves the pin instead of changing it. Switch to the Poke Tool first.

اردو میں سمجھیں

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CHECK YOURSELF

1. Which icon is the input pin: square or circle?
2. Which tool do you use to change an input between 0 and 1?

Answers: 1. Square (output pin is the circle). 2. The Poke Tool (pointing finger).

Next lecture: 1.2.1 – 1.2.3 Software Development Life Cycle (SDLC)